Primum Non Nocere? Mortality Effects of a Physician Strike in Germany

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Abstract

We study the consequences of a protracted and unanticipated physician strike in German university hospitals on patient mortality. To estimate the causal effect of the strike, we compare striking and non-striking hospitals in a difference-in-differences design using a comprehensive dataset of all hospital admissions in Germany linked to newly collected records of strike participation. Our results show that the strike led to a sharp decrease in hospital admissions and a significant increase in hospital mortality in striking hospitals. Applying a double selection LASSO algorithm to adjust for care rationing and patient selection, we show that around half of the mortality effect of the strike can be attributed to quality-of-care factors. Finally, we find spillover effects on admissions to nearby non-striking hospitals but no corresponding impact on hospital mortality. We conclude that prolonged industrial action in the healthcare sector may have severe consequences for patient safety unless appropriate risk mitigation protocols are established.

Keywords: physician strike, quality of care, rationing.

JEL Classifications: H75, I11, J52.

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1 Introduction

Labour disruptions by physicians are a controversial topic in public debates, as physicians' professional duty to provide patient care can come into conflict with their fundamental labour rights (Metcalfe, Chowdhury and Salim, 2015).[1] Even temporary disruptions in access to healthcare raise serious concerns about patient safety, the continuity of medical services, and the resilience of the healthcare system. However, despite considerable media attention and the ethical controversy surrounding physician strikes, there is limited causal evidence on how such disruptions affect patient health.

In this paper, we provide new evidence on this topic by exploiting an unexpected nationwide physician strike in Germany to study its causal effect on mortality. The strike followed a sudden breakdown in negotiations with the physicians' labour union and lasted three months in university hospitals in 2006.[2] It was one of the longest labour disruptions in German history, with about 2/3 of physicians in tertiary hospitals participating at its peak. Its scale and unexpected onset provide a unique natural experiment to identify the impact of a major disruption to healthcare supply.

The effect of a physician strike on patient health is not given a priori. On the one hand, labour disruptions can reduce quality of healthcare due to reduced access, higher workload among healthcare staff, and crowding effects. For example, Avdic (2016) shows that an increased distance to hospitals due to emergency department closures resulted in a lower probability of survival of an acute myocardial infarction in the short run, due to an increase in the risk of out-of-hospital mortality. Likewise, Piérard (2014), Lin (2014) and Aiken et al. (2002) highlight the importance of the staff-to-patient ratio in determining patients' outcomes. On the other hand, hospitals can respond by prioritizing urgent cases and postponing elective procedures, which could even improve short-term outcomes by reducing unnecessary treatment. In fact, overtreatment in hospitals is well documented (Lyu et al., 2017), suggesting that some disruptions could even have neutral or beneficial effects. As such, the net effect of a physician strike on patient health is ambiguous and an empirical

[1] The International Labour Organization (ILO) classifies hospitals as "essential services", meaning that strikes may be restricted or prohibited if they pose risks to life, safety, or public health (cf. International Labour Office, 2006).

[2] The majority of German tertiary hospitals are organised on federal state level and provide medical care to all publicly as well as privately insured individuals. They incorporate medical schools and are also involved in research and teaching activities.

question.

We address this question by combining comprehensive administrative data on hospital admissions and mortality with newly collected detailed information on hospital-level strike participation. We digitized newspaper reports to identify the exact start and end dates of the strike for each university hospital in Germany and matched this information to hospital records. To estimate the causal effects of the strike, we implement a difference-in-differences design that leverages variation in strike exposure across university hospitals, comparing outcomes in striking and non-striking hospitals before, during, and after the strike.

We begin our analysis by estimating the effect of the strike on in-hospital mortality and admissions. Our results suggest that the strike led to a sharp reduction in hospital admissions in striking hospitals of about 12% compared to non-striking hospitals, with no evidence of post-strike catch-up effects after the strike ended. The strike led to a significant increase in in-hospital mortality rates, with an estimated rise of about 0.16 percentage points, roughly a 9% increase relative to the baseline.

However, the composition of the patients may have changed during the strike, for example, if striking hospitals prioritized more urgent or frail patients. Although our estimated mortality effect captures the causal effect of the strike, it may not correspond to the policy-relevant effect if case-mix changed. To assess whether the change in mortality reflects a change in care quality or patient composition, we apply a double selection Lasso model (Belloni, Chernozhukov and Hansen, 2014) that accounts for observable differences in patient characteristics between treatment and control group. Once we account for strike-induced changes in patient characteristics, the coefficient on in-hospital mortality is reduced by half, but remains statistically significant. This suggests that the observed increase in mortality is partly driven by triaging strategies, where more urgent or frail patients are prioritized over healthier ones, and partly by reductions in care quality.

Although the covariate adjustment method ensures that our estimated mortality effect accounts for changes in patient case-mix, it does not yield interpretable coefficients for individual characteristics. Therefore, we also directly examine changes in patient composition. We show that patients admitted to striking hospitals were younger and more likely to be urgent cases. In line with this, emergency admissions increased by 5%, and the average length of stay increased by 0.2 days, possibly reflecting more intensive treatment needs in a population of sicker patients.

While these shifts suggest that some patients, likely those in better health, may have foregone hospital treatment during the strike, this does not appear to have led to excess mortality outside hospitals. Using comprehensive data on all deaths in Germany, we find no evidence of increased out-of-hospital mortality, indicating that the adverse effects of the strike were largely confined to inpatient care.

In the final part of the paper, we examine whether the strike caused spillover effects in nearby hospitals, as patients seeking care elsewhere may have strained their capacity and potentially affected quality of care. We find modest increases in admissions at nearby hospitals during the strike period but no corresponding rise in in-hospital mortality. Taken together, these results suggest that healthier patients at striking hospitals likely avoided care or were triaged, leading to a patient composition with a higher share of 'frail' patients and higher underlying mortality risk. Yet, our case-mix-adjusted mortality estimates suggest that care rationing can only explain half of the increase in-hospital mortality during the strike, with the remainder likely due to reductions in care quality.

We contribute to the scant literature on the causal effects of strikes in healthcare facilities. To our knowledge, only two papers have investigated the impact of physician strikes. Costa (2019) investigates the effects of multiple strikes caused by physicians, nurses, and diagnostic and therapeutic technicians in Portugal. While no effects for nurses' and diagnostic and therapeutic technicians' strikes can be found, physician strikes increased in-hospital mortality by about eight percent. Stoye and Warner (2023) analyse the impact of a number of short strikes of junior doctors in the United Kingdom in 2016. Even though our research question is related, our design is very different from both papers. Whereas they consider short and repeated disruptions caused by strikes, we study the impact of a prolonged strike by physicians that affected entire hospitals over a period of three months. Therefore, the shock we consider represents a more serious challenge to the health care system, but it also entails more time for the system to adapt to this disruption. In this sense, our analysis provides a strong test of the resilience of the system in the presence of a major unexpected challenge.

In terms of strikes of other healthcare workers, Gruber and Kleiner (2012) analyse multiple nurses' strikes in New York between 1984-2004, finding that strikes lead to an 18 percent increase in in-hospital mortality and lower quality of medical care. Kronborg, Sievertsen and Wüst (2016) and Hirani, Sievertsen and Wüst (2019) examine a nurses' strike in Denmark in 2008, showing reductions in prenatal midwife consultations, shorter hospital stays after birth, and fewer home visits.[3] Friedman and Keats (2019) study infant and neonatal health during a health worker strike in Kenya and find that babies born during strike months faced a higher risk of death. We contribute by providing evidence on physician strikes, a less frequently studied group whose responsibilities, such as diagnosis, treatment decisions, and surgeries, differ markedly from the care and monitoring typically provided by nurses and midwives. These differences in roles imply that physician strikes can disrupt healthcare systems in different ways.

The remainder of the paper proceeds as follows. Section 2 describes the setting. Section 3 gives information on the different datasets used in the analyses and section 4 describes our empirical strategy. Section 5 presents the results on the effects of the strike. Section 6 concludes.

2 Background

2.1 The German Hospital Sector

The German hospital sector treats about 19 million patients every year and involves approximately 2,000 hospitals with 500,000 beds (Federal Statistical Office, 2018). Hospitals are owned by public, private or non-profit organisations. Public hospitals can be separated further into hospitals run by municipalities, the 16 federal states, the state, and hospitals related to the statutory accident insurance (Goepfert and Conrad, 2013). The majority of the 36 university hospitals, which are the focus of our analysis, are run by the federal states. They are linked to the medical faculties of the universities and are thus also involved in research and teaching activities.

Roughly 10 percent of the 19 million patients treated in hospitals are admitted to university hospitals (Federal Statistical Office, 2018). Although they make up less than two percent

[3] They also show that missing earlier home visits due to the strike led to more frequent and urgent general practitioner visits in early childhood, compared to missing later visits.

of the hospitals, they disproportionally hold about 20 percent of the physicians working in hospitals.[4] As there is free choice of healthcare providers, patients can freely choose the hospital they want to be treated in and physicians in the primary or ambulatory care sector are not obliged to transfer patients to a specific hospital (de Cruppé and Geraedts, 2017). Treatment costs are usually covered by a patient's health insurance, as health insurance is mandatory in Germany and all legal residents have universal coverage by either public or private health insurance (Blümel and Busse, 2015).[5]

2.2 The Marburger Bund – Germany's Trade Union for Physicians

The *Marburger Bund* is a professional organisation and the trade union for physicians in Germany, covering about 70 percent of all physicians employed in hospitals (Greef, 2012). Before the physician strike in 2006, it acted primarily as a professional organisation and was represented in collective bargaining negotiations by the *Vereinte Dienstleistungsgewerkschaft (ver.di)*, a multi-branch trade union (Greef, 2012).

During negotiations for a new collective bargaining agreement for public services in 2005, the *Marburger Bund* demanded better working conditions for physicians working in hospitals (Marburger Bund Bundesverband, 2012). As *ver.di* was unwilling to negotiate a collective bargaining agreement specifically for physicians, the *Marburger Bund* decided to terminate their partnership with *ver.di* to become the first and only trade union for physicians.

2.3 Physician Strike

The *Marburger Bund* and the employer's association officially started collective bargaining for hospital physicians at the federal state level on October 12, 2005 (Greef, 2012). As the two parties were unable to reach an agreement, the largest physician strike in German history began on March 16, 2006 and continued until June 16, 2006, when a collective bargaining agreement was finally signed.

The strike, that took place in the majority of German university and federal hospitals,

[4] Table A.2 in Appendix A shows how the number of physicians and other health care workers in all German hospitals and university hospitals evolved between 2000 and 2010.

[5] The majority of patients (88%) are publicly insured and only about 11% are covered by a private health insurance (Bundesministerium für Gesundheit, 2019). The data used in this analysis covers privately and publicly insured patients. However, it is not possible to distinguish between privately and publicly insured.

was one of the longest labour disruptions in German history (Martens, 2008).[6] First, these labour disruptions were restricted to a few university and federal hospitals and to one or two days per week, but soon the strikes expanded to other university hospitals and lasted longer and longer; sometimes even a whole week. At the peak of the strike, about 2/3 of all physicians in these hospitals participated. In general, emergency treatment was ensured and also child birth units were excluded from the strike in most hospitals.[7]

Table 1 presents a chronology of events starting in 2005. The table also presents information on the strike in hospitals at the municipality level. However, the focus in this paper will be on the strike at the federal state level, as university hospitals are organised on the federal state level.

TABLE 1. Chronology of events

Date	Event
14 Apr 2005	ver.di and employer's association start bargaining (federal)
9 Sep 2005	MB terminates collective partnership with ver.di
12 Oct 2005	MB and employer's association start bargaining (federal)
9 Mar 2006	MB and employer's association start bargaining (local)
16 Mar 2006	Strike begins (federal)
16 Jun 2006	MB and employer's association reach agreement for physicians (federal)
26 Jun 2006	Strike begins (local)
17 Aug 2006	MB and employer's association reach agreement for physicians (local)

NOTE.— MB = Marburger Bund; ver.di = Vereinte Dienstleistungsgewerkschaft; Sources: Greef (2012) and Martens (2008).

3 Data

3.1 Strike Data

One challenge in studying the effects of the 2006 strike is that there is no reliable administrative data on strike participation that could be used for analysis (Dribbusch, 2018). Therefore, we collected information on the strike from newspaper articles published in 2006 to obtain

[6]There were 33 university hospitals in total, but only 24 of them participated in the strike. The university hospitals in Berlin, Frankfurt, Gießen and Marburg did not participate, as Berlin and Hesse are not part of the employer's association. Berlin was excluded from the employer's association in 1994 and Hesse left in 2004 (Greef, 2012). Hamburg and Schleswig-Holstein (Kiel and Lübeck) did not participate as they had reached a collective agreement before the strike took place. The University hospitals in Homburg, Bochum, Mannheim and Wuppertal did not participate in the physician strike for other reasons.

[7]This was settled by emergency agreements, signed by the university hospitals and the *Marburger Bund*.

information on the start and end date of the labour disruptions for each university hospital. In total, physicians from 24 of 33 university hospitals went on strike; nine of these participated from the first day onwards, while the remainder of the participating hospitals joined a few weeks later.[8] The remaining nine university hospitals were either not part of the collective bargaining agreement with the TdL in early 2006, or had already reached an agreement with their respective state before the strike began, and therefore did not participate in the industrial action.

One limitation of our strike data is that it captures only the extensive margin of participation in the strike, that is, whether and when a hospital joined the strike, but not the intensive margin, such as the number or share of physicians participating. Suggestive evidence indicates that the strike was more intense during its second month.

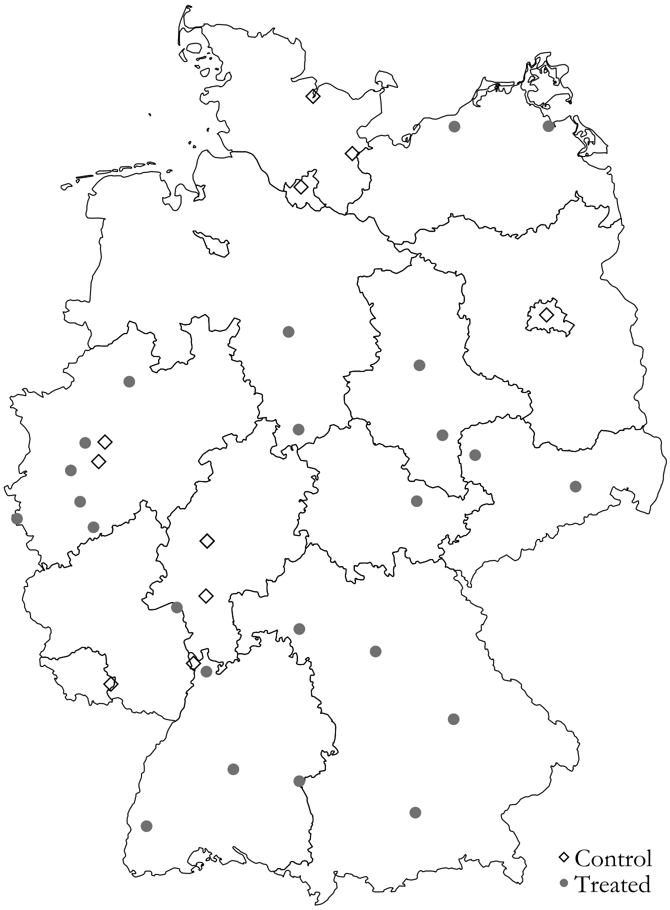
Descriptives Figure 1 shows descriptives for the strike. Panel (a) shows the location of treatment and control university hospitals in Germany, and panel (b) the distribution of strike durations across the striking hospitals. Panel (a) shows that treatment and control hospitals are located in different districts and usually even in different federal states, such that we are not concerned about any direct spillover effects of the strike on control hospitals. Panel (b) shows that the mean strike duration was approximately 83 days, with a median of 88 days, much longer compared to previous studies. For comparison, Gruber and Kleiner (2012) report a mean strike duration of only 32 days and a median of 19 days. This difference highlights that our setting involved substantially more prolonged disruptions.

3.2 Hospital-Level Data

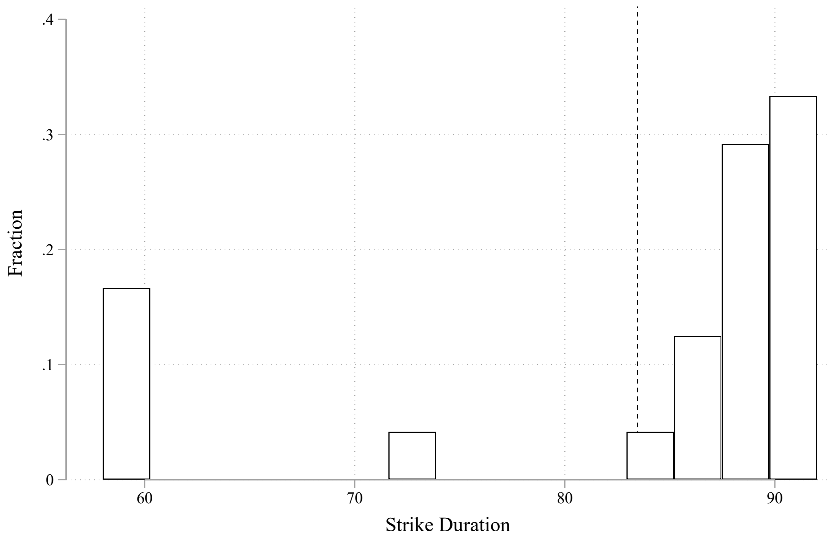
Hospital-level data on admissions for the years 2000 to 2008 was obtained from the German Federal Statistical Office (FDZ der Statistischen Ämter des Bundes und der Länder, 2000-2008*a*). It contains individual-level information on each hospital admission, including date of admission, length of stay, mortality and surgery indicators, 3-digit ICD-diagnosis, and patient characteristics such as age, gender, and place of residence. Hospital level characteristics, such as the number of physicians and nurses, are available yearly and used for balance tests (see Section 4). Since German hospitals are required to provide this information by law,

[8] Appendix Table A.1 shows the start and end dates of the strike for each participating university hospital.

FIGURE 1. Descriptives Strike



(a) Locations of treatment and control hospitals



(b) Strike duration

NOTE.— This figure shows the location of treatment and control hospitals in panel (a), and the duration of the strike across hospitals in panel (b). The dashed line in panel (b) indicates the average strike duration.

the data is complete and of high quality (see KHStatV, 1990).

Main Outcome Variables The primary outcome variable in this analysis is the daily mortality rate (patient deaths per 100 admissions) by hospital. Following Gruber and Kleiner (2012) and Geweke, Gowrisankaran and Town (2003), we also examine deaths occurring within 10 days of admission. This outcome helps mitigate potential bias from strike-induced changes in length of stay and better isolates mortality attributable to initial hospital care, excluding patients who may have been transferred elsewhere. In addition, we examine the daily number of deaths and daily number of admissions per hospital as outcomes.

To assess broader effects of the strike, we also examine whether it influenced hospital decisions and patient case-mix. To capture changes in case-mix, we consider patient age and gender, as well as indicators of illness severity derived from detailed diagnosis codes. Specifically, we use an ambulatory care sensitive indicator (ACS) based on Sundmacher et al. (2015)[9] and an emergency and urgency indicator based on the classification by Krämer, Schreyögg and Busse (2019).[10] We also study effects on the length of stay (in days) and surgery rate (surgeries per 100 admissions). Changes in length of stay can reflect capacity constraints, earlier discharges, or adjustments in care intensity, while changes in the surgery rates can indicate changes in prioritisation, such as postponement of elective procedures or changes in the types of patients admitted during the strike.

Sample and Descriptive Statistics For the main regressions on the hospital level, we keep only university hospitals as they provide a balanced sample, whereas other hospitals could have been exposed to the strike on the municipality level starting in June 2006, for which we do not have strike data. We aggregate the data on the day of admission and hospital level. The final sample consists of 143,744 observations.

Exposure to the strike is defined as being admitted to a striking hospital during the strike period. The variable takes on the values zero or one. This variable does not capture patients that already were hospitalised when the strike began. We therefore use an alternative

[9] The ACS indicator helps to identify hospitalisations which could have been avoided by effective and timely ambulatory care (Sundmacher et al., 2015).

[10] Krämer, Schreyögg and Busse (2019) developed a diagnosis-based classification to determine whether a hospitalisation is elective or an emergency. They used machine learning and several predictor variables to classify the different diagnosis. A hospitalisation is defined as an emergency if their urgency indicator is above 0.5.

treatment indicator in a robustness check, defined as the share of patients on a specific day that were affected by the strike at any time during their hospital stay (see Section 5.5). We prefer our main treatment definition because it avoids bias from the potential endogenous impact of the strike on length of stay. Table 2 presents descriptive statistics for the main outcome variables, treatment indicator, and patient characteristics by the treatment status of the hospital over all observation years.

TABLE 2. Descriptive statistics: Hospital data

	Treatment Hospital			Control Hospital		
	Mean	SD	Obs.	Mean	SD	Obs.
Treatment status	0.03	0.16	84,986	0.00	0.00	58,758
Female	0.49	0.05	84,986	0.49	0.07	58,758
Age	47.38	3.89	84,986	48.99	5.28	58,758
Hospital daily cases	159.28	69.88	84,986	156.27	111.23	58,758
Hospital log daily cases	4.95	0.54	84,986	4.79	0.81	58,758
Length of Stay	8.12	1.59	84,986	8.26	2.13	58,758
Surgery rate	37.06	22.17	84,986	33.57	22.03	58,758
Hospital log daily mortality	0.83	0.59	68,985	0.92	0.66	38,410
Mortality rate	1.67	1.45	84,986	1.94	2.17	58,758
10-Day mortality rate	0.91	1.11	84,986	1.08	1.64	58,758
Emergency rate	27.65	12.18	84,648	30.03	13.22	58,508
Urgency	0.33	0.09	84,648	0.35	0.09	58,508

NOTE.—Descriptive statistics by treatment status of the hospital. Treated hospitals are university hospitals that participated in the strike and control hospitals are those that did not participate. Hospitals are weighted by pre-strike admissions. Variable definitions are available in Appendix B. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

3.3 Mortality Census

Mortality data was also obtained from the German Federal Statistical Office and includes all deaths (both in and out of hospitals) and causes for death (3-digit ICD-Codes) between 2000 and 2008. The district mortality data is useful to study effects on mortality outside of the hospital. For example, it could be that patients affected by the strike died outside of the hospital if they were discharged too early or failed to be admitted in time for a critical condition.

Sample and Descriptive Statistics We use district of residence and date of death to merge the mortality data with the collected strike information and subsequently collapse the data into district-date cells.[11] Treatment assignment is based on district of residence at the time of death. Treated districts are those with a striking university hospital, while control districts are those without a striking university hospital. Our final analysis data sample consists of approximately 80,000 district-date cells which we use as observations in our regressions. Descriptive statistics for the mortality data are presented in Table 3.

TABLE 3. Descriptive statistics: Mortality data

	Treatment District			Control District		
	Mean	SD	Obs.	Mean	SD	Obs.
Treatment variable	0.04	0.19	56,837	0.00	0.00	23,404
Female	0.55	0.19	56,837	0.55	0.12	23,404
Age at death	75.70	5.83	56,837	75.41	3.57	23,404
Mortality rate (per 10,000)	0.26	0.10	56,837	0.28	0.07	23,404

NOTE.—Descriptive statistics by treatment status of the district. Treated districts are those with a university hospital that participated in the strike and control districts are those without. Districts are weighted by pre-strike population. Variable definitions are provided in Appendix B. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*b*), own calculations.

4 Econometric Strategy

We employ several complementary approaches to estimate the effects of the strike and examine the underlying mechanisms driving the observed mortality effect. In a first step, we use a difference-in-differences (DiD) design comparing 24 striking university hospitals to nine non-striking ones. Second, we apply a similar DiD approach at the district level, comparing districts with striking university hospitals to those with non-striking hospitals. This allows us to examine whether the strike affected overall mortality, including deaths outside the hospital. Third, to assess whether changes in mortality are primarily driven by changes in patient composition during the strike, we combine a a double-selection Lasso approach (Belloni, Chernozhukov and Hansen, 2014) with our main DiD analysis. This allows us to flexibly control for a large set of patient characteristics and isolate the mortality effect net of strike-induced case-mix changes. Finally, we analyze whether patients shifted to nearby

[11]The district (*Kreis*) is the intermediate administrative level between municipality (*Gemeinde*) and federal state (*Bundesland*). These units have responsibilities in the domains of public transport, road construction, and the construction of hospitals. There were in total 323 of them in 2006.

non-striking hospitals by exploiting variation in pre-strike patient shares across districts. We describe each analysis in more detail below.

4.1 Hospital Level Effects

We begin our analysis by examining the effects of the strike on hospital admissions and in-hospital mortality. To this end, we compare outcomes before, during, and after the strike between university hospitals that were affected by the strike and those that were not. Our main regression specification is:

$$y_{it} = \beta_0 + \beta_1 S_{it} + \mu_i + \nu_t + \gamma_H (H_i \cdot t) + \varepsilon_{it} \tag{1}$$

where y_{it} is one of the outcome variables listed in Table 2 for hospital i on day t, S_{it} is an indicator variable equal to one for hospitals that participated in the strike during the strike period and 0 otherwise, μ_i are hospital fixed effects, ν_t are date of admission fixed effects and $H_i \cdot t$ is a vector of hospital-specific linear time trends. Date of admission fixed effects include fixed effects for year, week, and day of week. All regressions are weighted by the baseline average admissions per day in each hospital and standard errors are clustered at the hospital level.

This analysis exploits variation in strike exposure across hospitals and over time. Hospital fixed effects control for time-invariant characteristics such as average case volume, number of beds, and the share of specialists. Date fixed effects account for common temporal trends, such as lower admission volumes on weekends. Including hospital-specific linear trends allows for differential outcome trajectories across hospitals. We also present results from a more parsimonious specification that excludes hospital-specific linear time trends. The key parameter of interest is β_1, which measures the average treatment effect on the treated (ATT) of the strike on the outcomes.

We also estimate event study models that allow us to use the months leading up to the strike to test for any differential pre-trends and to examine time-varying treatment effects. Specifically, we estimate an event study specification using two-month bins from January/February 2005 to January/February 2007. The estimating equation is:

$$y_{it} = \alpha_0 + \sum_{\substack{k=\text{Jan/Feb 2005} \\ k\neq\text{Jan/Feb 2006}}}^{\text{Jan/Feb 2007}} \beta_t \left(\mathbf{1}[t=k] \cdot S_{it}\right) + \sum_{\substack{k=\text{Jan/Feb 2005} \\ k\neq\text{Jan/Feb 2006}}}^{\text{Jan/Feb 2007}} \gamma_k \left(\mathbf{1}[t=k]\right) + \mu_i + \nu_t + \varepsilon_{it} \tag{2}$$

where y_{it} denotes the outcome for hospital i on day t, $\mathbf{1}[t=k]$ is a set of indicator variables for two-month bins k relative to the two-month bin before the strike ($k =$ Jan/Feb 2006). We include hospital fixed effects μ_i and date fixed effects ν_t. Standard errors are clustered at the hospital level, and all regressions are weighted by baseline average daily admissions in each hospital.

Identification Assumption Our identification strategy relies on the assumption that, in the absence of the strike, treated and control hospitals would have followed parallel trends in outcomes (conditional on the included controls). We assess the plausibility of this assumption in two ways. First, we use event study estimates to examine whether there are any differential pre-trends between treated and control hospitals. Second, we test for balance in key hospital characteristics between the two groups.

Table A.3 presents hospital balancing tests to study the comparability between treated and untreated hospitals in our sample.[12] To study this, we regress each of the variables reported in the table on a set of control variables and the treatment variable indicator. The point estimates from Column (1), where we only include year fixed effects as controls, suggest that treated hospitals are on average larger with greater number of beds, more staff, and a higher share of intensive care unit beds. However, once we additionally control for hospital fixed effects and linear trends, the two groups are well balanced across all observable characteristics. We also consider an alternative control group as a robustness check.

4.2 District Level Effects

In addition to in-hospital mortality, we investigate aggregate district-level mortality effects by estimating the following event study specification:

[12] As the data on hospital level characteristics is only available yearly, we coded the entire year 2006 as treatment period.

$$
\begin{aligned}
y_{jt} = \delta_0 &+ \sum_{\substack{k=\text{Jan/Feb 2005} \\ k\neq \text{Jan/Feb 2006}}}^{\text{May/Jun 2006}} \delta_k \left(\mathbf{1}[t=k] \cdot S_{jt}\right) \\
&+ \sum_{\substack{k=\text{Jan/Feb 2005} \\ k\neq \text{Jan/Feb 2006}}}^{\text{May/Jun 2006}} \theta_k \mathbf{1}[t=k] + \mu_j + \nu_t + \gamma X_{jt} + \varepsilon_{jt}
\end{aligned}
\tag{3}
$$

where y_{jt} denotes the mortality rate (per 10,000 inhabitants) in district j on day t, S_{jt} is an indicator equal to one if the district includes a hospital affected by the strike during the strike period, $\mathbf{1}[t=k]$ is a set of indicator variables for two-month bins relative to the two-month bin before the strike ($k =$ Jan/Feb 2006).[13] The model includes district fixed effects μ_j and date fixed effects ν_t, as well as controls for time-varying district-level characteristics X_{jt}, such as the share of females in the district population. Standard errors are clustered at the district level, and regressions are weighted by baseline district population.

The key parameters of interest are the δ_k, which measure the impact of the strike on district mortality relative to the two-month bin before the strike.

4.3 Accounting for Patient Selection in Mortality Estimates

It is important to recognise that the strike may have changed the patient composition. While emergency care was still available, elective surgeries and other non-emergent cases might have been affected. If emergency care patients have worse health outcomes than non-emergency care patients, this change in the case-mix will contribute to our DID estimates. Whereas the estimated effect captures the causal effect of the strike on in-hospital mortality rates, it may not correspond to the policy-relevant causal effect.

To infer whether quality of care was compromised during the strike, we need to base estimates of mortality effects on a comparable population of patients. We address this concern by controlling for patient composition in the empirical analysis using a two-step lasso shrinkage technique introduced by Belloni, Chernozhukov and Hansen (2014).[14]

[13] We estimate the effects of the strike on district mortality only until May/June 2006, as another strike occurred at the district level thereafter and data for this subsequent strike are not available.

[14] We also estimate the effects of the strike on district-level mortality rates (see Section 4.2) to examine whether out-of-hospital mortality increased due to compromised quality of care - for example, if patients were discharged too early or failed to be admitted in time for a critical condition.

To account for strike-induced changes in patient composition, we combine the various patient and case characteristics that are available in the data – age, gender, detailed ICD-10 codes, indicators of urgency, and deferability – there are 337 specific covariates that may be considered. We rely on a double selection Lasso model, where covariates that predict either the treatment (being admitted to a striking hospital in the strike period), or the outcome variable, are included in the specification (Belloni, Chernozhukov and Hansen, 2014). To avoid over-fitting, we impose a penalty on the number of predictors, where the tuning parameter has been chosen by five-fold cross-validation. We add the selected variables as control variables to the main regression specification (1).[15] The main coefficient of interest, β_1, then yields an estimate of the effect of the strike on in-hospital mortality, net of strike-induced changes in patient composition, thereby providing a more policy-relevant measure of the impact of the strike on the quality of care.

4.4 Analysis of Spillover Effects

If admissions in striking hospitals decrease and only more fragile patients are admitted, healthier patients may decide to either delay hospitalisation, seek treatment at an other hospital or substitute other types of care for hospital care. Our estimates of post-strike admissions and mortality will give an indication of whether postponement is a common strategy. In what follows, we also analyse the extent to which surrounding non-university hospitals were affected by the strike.

We use the year 2005 as our benchmark to calculate, for each district, the share of patients who receive treatment in any strike-hit university hospital in that year. We argue that this variable is a good proxy to study strike spillover effects, since the pressure on surrounding non-treated hospitals should be greatest in districts where a high share of patients was treated in striking hospitals prior to the strike. The pre-strike patient shares is used as a regressor in the following regression model:

$$y_{it} = \beta_0 + \beta_1 S_{it} + \mu_i + \nu_t + D_i \cdot t + \varepsilon_{it} \tag{5}$$

[15] Specifically, we estimate the following specification:

$$y_{it} = \beta_0 + \beta_1 S_{it} + \mu_i + \nu_t + \gamma_H (H_i \cdot t) + \delta X_{it} + \varepsilon_{it}, \tag{4}$$

where all variables are defined as before (see section 4.1) and X_{it} are the selected variables.

where y_{it} is a specific hospital outcome for district i on day t. S_{it} is the interaction between a strike period indicator and the pre-strike share of patients treated in striking university hospitals. The remaining variables are defined as before.

5 Results

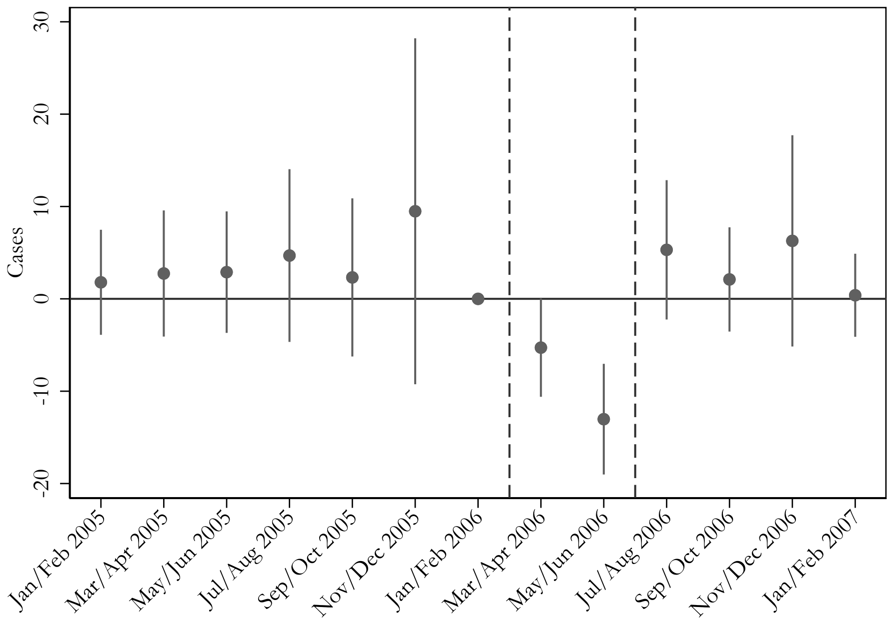
5.1 Hospital Admissions

We first investigate whether the 2006 physician strike affected the number of daily hospital admissions. Figure 2 illustrates bi-monthly event study estimates for the relative change in hospital admissions for striking and non-striking hospitals, estimating equation (2). The absence of a trend for the point estimates in the time periods leading up to the strike suggests that the critical common trend assumption required for a causal interpretation of our DID estimator is fulfilled.

Considering the estimates for the strike period itself, indicated by the vertical dashed lines, we observe a significant relative drop in admissions in striking hospitals, in particular in May and June 2006 when the strike was at its most intense phase. The coefficient estimates again turn statistically insignificant in the months immediately following the strike, suggesting that the strike caused a net overall drop in the number of admissions.[16]

[16] Appendix figure A.1 shows that this decrease is driven by a decrease in admissions in treated hospitals rather than any change in admissions in the control group.

FIGURE 2. Event study results: Hospital admissions



NOTE.— The figure shows event study estimates for admissions, controlling for day of week, week, year, and hospital fixed effects. The vertical lines represent the beginning and end of the strike. 95% confidence intervals included. Robust standard errors are clustered at the hospital level. Regressions are weighted by number of pre-strike admissions. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

Formal DID estimates are reported in Table 4, confirming the event study result that the strike significantly decreased hospital admissions during the strike period. In particular, when including time and hospital fixed effects and hospital-specific linear time trends, the estimates from Columns (1) and (2) suggest a decrease in daily admissions by 19-21 cases on average, or 12 percent.

TABLE 4. Difference-in-differences estimates: Hospital admissions

	Observations	Mean	(1)	(2)
Cases	143,744	158.32	-21.460***	-19.041***
			(4.646)	(2.596)
Log Cases	143,744	4.90	-0.122***	-0.118***
			(0.019)	(0.014)
Year FE			✓	✓
Hospital FE			✓	✓
Hospital Specific Trends				✓

NOTE.— Robust standard errors clustered at the hospital level are reported in parenthesis. Significance levels: $*$ 0.10 $**$ 0.05 $***$ 0.01. Regressions include day of week, week, year, and hospital fixed effects, and hospital specific time trends. Regressions are weighted by number of pre-strike admissions. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

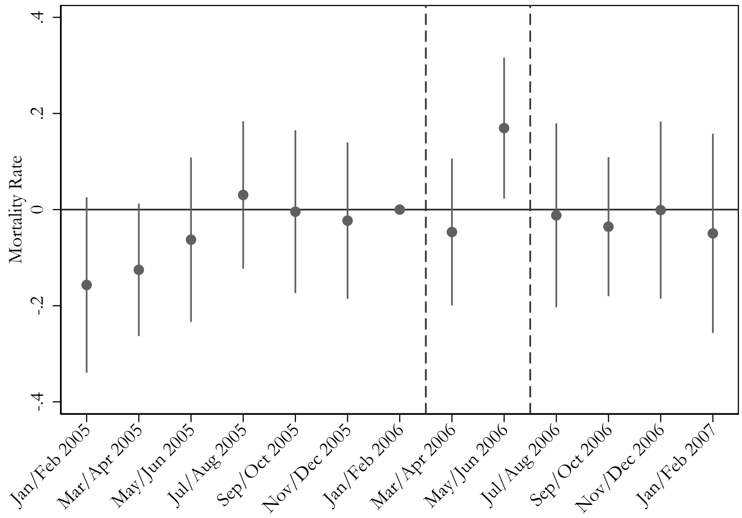
5.2 Mortality

In-hospital Mortality Next, we analyse the consequences of the strike on in-hospital mortality rates. Figure 3 presents event study graphs for the mortality rate and 10-day mortality rate. The latter helps address potential bias from strike-induced changes in length of stay and better isolates deaths attributable to the initial phase of hospital care, excluding patients who may have been transferred elsewhere. Again we observe that striking and non-striking hospitals had comparable mortality trends in the lead-up to the strike, supporting our empirical strategy.

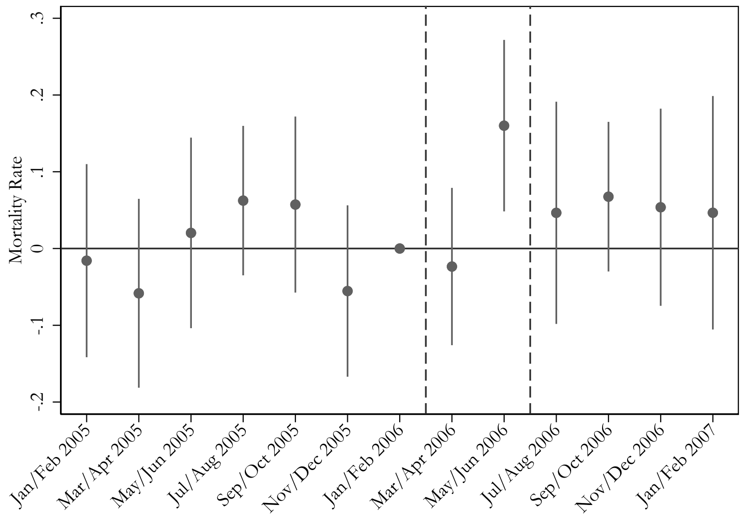
We also see a reverse trend in the estimates for the strike months with a relative increase in both in-hospital and 10-day mortality rates for striking hospitals during the intensive strike period in May and June. These estimates return to being close to zero and statistically insignificant at the end of the strike period.[17]

[17] Appendix figure A.2 shows that this increase in in-hospital mortality is driven by an increase in in-hospital mortality of treated hospitals.

FIGURE 3. Event study results: Hospital mortality



(a) In-hospital Mortality Rate



(b) 10-Day In-hospital Mortality Rate

NOTE.— The figures show event study estimates for the mortality rate and 10-day mortality rate, controlling for day of week, week, year, and hospital fixed effects. The vertical lines represent the beginning and end of the strike. 95% confidence intervals included. Robust standard errors are clustered at the hospital level. Regressions are weighted by number of pre-strike admissions. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

DID estimates corresponding to the event study plots for the mortality outcomes are presented in Columns (1) and (2) of Table 5. The point estimates imply that the mortality rate increased by roughly 0.16 percentage points during the strike period, corresponding to an increase of around nine percent compared to the baseline mean of 1.75 deaths per 100 admissions. The estimates for 10-day mortality are somewhat lower at 0.07 percentage points, or

7.3 percent, but remain significant at the five percent level of statistical significance. Finally, reported estimates from the third row of the table indicate that the number of in-hospital deaths drops by 0.07 from a baseline of 2.46; a reduction by 2.8 percent. This estimate is not significantly significant, yet consistent with the estimates for admissions (-12%) and mortality (+9%) since $0.88 \cdot 1.09 = 0.96$.

Accounting for Patient Selection in In-hospital Mortality Effect While the estimated mortality effect reflects the causal effect of the strike, it does not necessarily represent a policy-relevant parameter if it is driven by changes in patient composition. To isolate the effect of the strike on in-hospital mortality from strike-induced changes in the composition of admitted patients, we apply the covariate adjustment approach described in Section 4.3. Columns (3) and (4) of the table report the results. The results suggest that, conditional on the selected covariates, the strike had a moderate effect on mortality rates: the increase is down to 0.7 percentage points or 4.1 percent. This suggests that part of the above estimated mortality effect is due to selection of patients in poorer health being more likely to present at striking hospitals during the strike period, and part by reductions in care quality.

TABLE 5. Difference-in-differences estimates: Hospital mortality

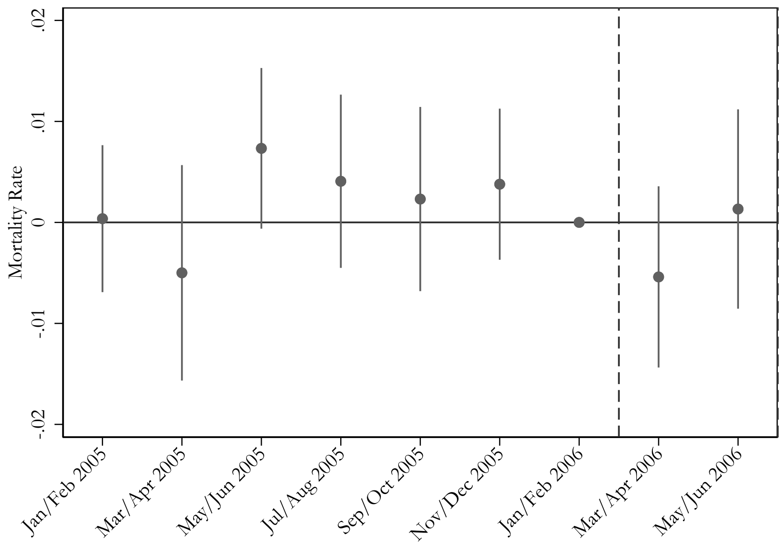
			Baseline Specification		Covariate Adjusted	
	Observations	Mean	(1)	(2)	(3)	(4)
Mortality Rate	143,744	1.75	0.159***	0.165***	0.069*	0.072**
			(0.043)	(0.041)	(0.037)	(0.035)
10-Day Mortality Rate	143,744	0.96	0.074**	0.073**	0.012	0.011
			(0.036)	(0.035)	(0.030)	(0.029)
Number of Deaths	143,744	2.46	-0.099	-0.069	-0.093	-0.068
			(0.087)	(0.056)	(0.076)	(0.054)
Predicted Mortality Rate	143,744	1.81	0.000	0.000		
		(0.000)	(0.000)			
Covariate adjustment					✓	✓
Year FE			✓	✓	✓	✓
Hospital FE			✓	✓	✓	✓
Hospital Specific Trends				✓		✓

NOTE.— Robust standard errors clustered at the hospital level are reported in parenthesis. Significance levels: $*$ 0.10 $**$ 0.05 $***$ 0.01. Regressions include day of week, week, year, and hospital fixed effects, and hospital specific time trends. Regressions are weighted by number of pre-strike admissions. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

Mortality Outside of Hospitals If the strike had significant and notable effects on patient outcomes, we would expect to observe at least some impact on overall district-level mortality. District mortality data is useful for detecting potential effects on deaths that occur outside hospitals, for example, if patients were discharged too early or were not admitted in time for a critical condition. However, since only 2.3% of all deaths occurred in university hospitals in the year before the strike, any such effect on overall mortality is likely to be small.

Figure 4 presents event study estimates comparing the out-of-hospital mortality rate (per 10,000 residents) in treatment and control districts, estimating equation (3). Treatment districts are defined as those with a striking university hospital, while control districts are those without a striking university hospital. The graph shows no significant increase in mortality during the strike period, suggesting that the strike had no larger spillover effects on mortality beyond the hospital setting.

FIGURE 4. Event study results: District mortality



NOTE.— The figure shows event study estimates for the district mortality rate (per 10,000 residents), controlling for day of week, week, year, and district fixed effects. The vertical lines represent the beginning and end of the strike. 95% confidence intervals included. Robust standard errors are clustered at the district level. Regressions are weighted by district population. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

5.3 Care Rationing

In the previous section, we used a covariate selection approach to flexibly account for strike-induced changes in patient composition when estimating the effect of the strike on in-hospital mortality. Although this approach ensures that the mortality effect we estimate accounts for changes in case-mix, it does not provide interpretable estimates for individual variables, making it difficult to assess how patient characteristics or hospital behaviour changed. In this section, we examine these changes directly.

Table 6 reports estimates for a set of outcomes associated with patient characteristics and hospital decisions, including age, sex, rates of Ambulatory Case Sensitive (ACS) and emergency cases, an indicator of urgency, a deferability index, length of stay and rate of surgery. Although some of these outcomes reflect pre-admission patient characteristics, others may be influenced by hospital behavior, and for many, the distinction is not clear cut.

The estimates presented in the table suggest that the strike led to a significant reduction in the age of the patients by 0.41 years and significant increases in both emergency rates and the probability that a patient is classified as an urgent case. This is consistent with the fact that emergency care patients in general tend to be younger on average.[18] We also estimate an increase in length of stay by 0.24 days, but we do not find significant changes in the composition of gender or ACS conditions. For surgery rates, we find a considerable, albeit only marginally significant, reduction by about 4.5 percent, suggesting a possible postponement of scheduled surgeries during the strike period. Together, these findings suggest that patients with relatively poorer health conditions were admitted to striking hospitals during the strike period.

[18]Figure A.3 in Appendix A shows the age distribution of non-emergency versus emergency care patients.

TABLE 6. Difference-in-differences estimates: Patient composition and hospital decisions

	Observations	Mean	(1)	(2)
		PATIENT COMPOSITION		
Age	143,744	47.89	-0.369**	-0.411***
			(0.157)	(0.146)
Female	143,744	0.49	0.002	0.002
			(0.002)	(0.002)
ACS Rate	143,744	20.30	-0.188	-0.301
			(0.221)	(0.183)
Emergency Rate	143,156	28.41	1.558***	1.505***
			(0.349)	(0.348)
Urgency Indicator	143,156	0.34	0.012***	0.012***
			(0.003)	(0.002)
Deferability Index	143,549	0.12	0.003***	0.003***
			(0.001)	(0.001)
		HOSPITAL DECISIONS		
Length of Stay	143,744	8.17	0.255***	0.243***
			(0.055)	(0.052)
Surgery Rate	143,744	35.95	0.016	-1.629*
			(1.409)	(0.930)
Year FE			✓	✓
Hospital FE			✓	✓
Hospital Specific Trends				✓

NOTE.— Robust standard errors clustered at the hospital level are reported in parenthesis. Significance levels: $*$ 0.10 $**$ 0.05 $***$ 0.01. Regressions include day of week, week, year, and hospital fixed effects, and hospital specific time trends. Regressions are weighted by number of pre-strike admissions. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

To further understand how the strike affected different patient groups, we split the sample by emergency status. This allows us to examine whether stronger strike effects are observed for non-emergency cases, as these patients may delay treatment or seek care at other facilities, whereas emergency cases generally require immediate treatment.

Table 7 presents separate DID estimates for emergency and non-emergency patients in our sample. Indeed, we see that the decrease in the number of admissions is considerably greater in magnitude for non-emergency cases (15.5%) compared to emergency cases (7.1%) suggesting a relative shift to more urgent cases. In addition, effects on mortality rates are stronger (relative to baseline means) and only significant for non-emergency admissions, and the impact on length of stay is also greater in magnitude for non-emergencies. Overall, the results in Table 7 suggest that healthier patients were less likely to receive care in striking hospitals compared to patients with poorer health status and more acute health conditions. Table A.4 presents the effects by quartiles of the deferability index, providing further support for this interpretation.[19]

TABLE 7. Difference-in-differences estimates: Heterogeneity by emergency status

	EMERGENCY			NON-EMERGENCY		
	Mean	(1)	(2)	Mean	(1)	(2)
Cases	143.06	-18.516***	-16.631***	164.18	-21.976***	-19.793***
		(4.197)	(2.294)		(4.640)	(2.700)
Log Cases	3.48	-0.069***	-0.071***	4.57	-0.159***	-0.155***
		(0.016)	(0.010)		(0.023)	(0.020)
Mortality Rate	3.37	0.133	0.143	1.27	0.207***	0.207***
		(0.102)	(0.098)		(0.044)	(0.043)
10-day Mortality Rate	2.21	0.071	0.072	0.55	0.080***	0.074***
		(0.080)	(0.079)		(0.026)	(0.025)
Number of Deaths	1.19	-0.064	-0.055	1.30	-0.029	-0.015
		(0.049)	(0.038)		(0.046)	(0.033)
Length of Stay	9.31	0.241**	0.232**	8.00	0.368***	0.358***
		(0.107)	(0.092)		(0.061)	(0.057)
Surgery Rate	24.10	0.047	-1.256	40.31	0.074	-1.798*
		(1.220)	(0.786)		(1.591)	(1.063)
Year FE		✓	✓		✓	✓
Hospital FE		✓	✓		✓	✓
Hospital Specific Trends			✓			✓

NOTE.— Robust standard errors clustered at the hospital level are reported in parenthesis. Significance levels: $*$ 0.10 $**$ 0.05 $***$ 0.01. Regressions include day of week, week, year, and hospital fixed effects, and hospital specific time trends. Regressions are weighted by number of pre-strike admissions. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

5.4 Spillovers to Nearby Hospitals

To study spillover effects to other (non-university) hospitals, we estimate equation (5) for a set of outcomes. Table 8 reports the estimated δ_1 parameter, interpreted as the average percentage change in the outcome variable for each percentage point of a hospital's pre-strike patient population treated in a striking hospital. The results indicate that hospitals surrounding treated university hospitals experience an increase in admissions by about eight percent.[20] However, there is no indication that this increase in admissions worsens patient

[19]The deferability index is defined following Card, Dobkin and Maestas (2009) and Halla, Liu and Liu (2019) and is explained in more detail in Appendix B.

[20]These spillovers are observed at non-university hospitals, which are excluded from the main analysis comparing treated to non-treated university hospitals. This ensures that the control group is not affected by

outcomes, as the parameter estimates for mortality rates and length of stay are generally small and non-significant at conventional levels.

TABLE 8. Difference-in-differences estimates: Spillover analysis

	Observations	Mean	(1)	(2)
Log Cases	1,041,569	4.25	0.102***	0.078***
			(0.031)	(0.020)
Mortality Rate	1,041,569	2.57	-0.118	0.039
			(0.124)	(0.094)
10-Day Mortality Rate	1,041,569	1.62	-0.070	-0.014
			(0.090)	(0.081)
Length of Stay	1,041,736	8.19	0.483***	0.105
			(0.143)	(0.109)
Year FE			✓	✓
District FE			✓	✓
District Specific Trends				✓

NOTE.— Robust standard errors clustered at the district level are reported in parenthesis. Significance levels: $*$ 0.10 $**$ 0.05 $***$ 0.01. Regressions include day of week, week, year, and district fixed effects, and district specific time trends. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

5.5 Robustness Checks

Placebo Event In this section, we examine a number of robustness checks to probe the stability of our empirical findings. First, to analyse whether seasonality is a confounding factor, we estimate hospital level placebo regressions by assigning a fake strike to the same period in 2004, a year with no actual strike activity. Table A.5 presents the results showing small and statistically insignificant estimates across the board.

Alternative Control Group Next, we test the stability of our findings to the choice of control group. To this end, we use all other hospitals in Hamburg and Berlin, who participated neither in the federal nor in the municipality strike, as alternative controls.[21] Note that this alternative control group includes all hospitals in the two federal states, not only university hospitals, and is therefore potentially less comparable with respect to a range of

spillovers.

[21]Berlin did not participate in the strike, because it was not a member of the employers' association and Hamburg reached a collective agreement before the strike began.

factors, including size and scope of services provided. Nevertheless, Table A.6 shows that our findings are largely robust to the choice of control group.

Alternative Treatment Indicator We also test the robustness of our main findings by specifying an alternative treatment indicator, as the one we use in our main analysis does not capture patients who were already hospitalised when the strike started. The alternative treatment indicator is defined as the share of patients on a specific day who were affected by the strike at any time during their hospital stay. The reported estimates from Table A.7 show that our results are robust to using this alternative definition of treatment group. We prefer our main treatment definition, as it is not influenced by any strike-induced changes in the length of stay.

Pre-trends Test We also test for linear pre-trends by estimating the following regression model on the pre-strike period:

$$y = \beta(trend \times treated) + \gamma treated + \delta trend + \varepsilon, \tag{6}$$

where $trend$ is a non-parametric time trend estimated for each $year \times month$ cell in the sample, and $treated$ is a treatment indicator equal to one for striking hospitals and zero for non-striking hospitals.

If the estimate β is significantly different from zero, this would suggest that treated and control hospitals had significantly different trends leading up to the strike; thus rejecting the common trend assumption required for consistency of the DID estimator. The reported $\hat{\beta}$ in Table A.8 indicate that the common trend assumption cannot be rejected at any conventional levels of statistical significance.[22] In addition, it should also be noted that our main specification includes hospital-specific time trends which makes differential pre-trends less likely to be a problem.

Staggered Treatment Adoption Finally, recent literature suggests that event study estimates may be biased if the timing of treatment is staggered and treatment effects are heterogeneous or evolve over time (Sun and Abraham, 2021; Goodman, 2014). In our setting,

[22] While recent work cautions that pre-trend tests may lack power (Roth et al., 2022), they still provide additional supporting evidence for the validity of the common trends assumption.

hospitals did not begin to strike on the same date, meaning our control group includes both never-treated and later treated hospitals. To show the robustness of our main results, we use the estimator proposed by Sun and Abraham (2021).[23] Figures A.4 and A.5 show that the results remain similar to those of our main specification.

6 Conclusion

Strikes in essential sectors, such as transport, healthcare and public safety, have become more common in recent times due to budget pressures, worker dissatisfaction with rationalisation, and welfare retrenchment policies in the public sector. In particular, in the healthcare sector, these factors have prompted instances of industrial action, potentially leading to both social and individual costs due to lower access to healthcare and lower quality of care. Given the critical nature of the services provided by medical professionals, it is therefore an important task to study the impact of strikes in the healthcare sector.

We explore the impact of a nationwide physician strike in German university hospitals in 2006 on patient mortality using a DiD design. We compare changes in outcomes over time between hospitals affected by the strike and those that were not. Our results show that admissions in striking hospitals dropped by 12 percent during the strike period compared to non-striking hospitals. In-hospital mortality rate increased by nine percent during the same period. However, this effect is reduced by half after adjusting for patient case-mix, suggesting that the observed increase in mortality is partly driven by changes in patient composition, where more urgent or frailer patients were prioritized, and partly by reductions in care quality. We find no evidence of delayed admissions after the strike or increases in out-of-hospital mortality. However, we observe modest spillovers in admissions to nearby hospitals, without corresponding effects on mortality.

Our paper provides additional nuance to the existing literature on strikes in the healthcare sector by focusing on a protracted and intensive strike, thus providing new insights beyond the typically shorter and recurring strikes studied in previous work. We interpret our findings as showing that strikes in healthcare (and elsewhere) can have very different impacts on organisational efficacy depending on the extent and range of services affected by

[23] We use monthly bins when applying the estimator of Sun and Abraham (2021), since treatment is staggered at the monthly level but not within months.

the disruption. However, one drawback of the analysis in this paper is that death is a rare and serious outcome. It is possible, and even likely, that other quality or patient care indicators, such as readmission rates or patient satisfaction, were similarly or more affected by the strike. There might also exist other economic consequences of the strike, such as financial losses for the hospitals or an increased workload for non-striking physicians, that we leave unexplored in this paper. Given the findings of this paper, establishing the total impact of strikes on a fuller set of social, economic, and health indicators could therefore constitute a fruitful avenue for further research.

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Appendix A Additional tables and figures

TABLE A.1. Start and End Dates of Striking University Hospitals

University Hospital	District Code	Federal State	Start Date	End Date
Tübingen	416	Baden-Württemberg	20/03/2006	16/06/2006
Heidelberg	221	Baden-Württemberg	16/03/2006	16/06/2006
Ulm	421	Baden-Württemberg	20/03/2006	16/06/2006
Freiburg	311	Baden-Württemberg	16/03/2006	16/06/2006
München	162	Bayern	16/03/2006	16/06/2006
Erlangen-Nürnberg	562	Bayern	20/03/2006	16/06/2006
Würzburg	663	Bayern	16/03/2006	16/06/2006
Regensburg	362	Bayern	20/03/2006	16/06/2006
Greifswald	001	Mecklenburg-Vorpommern	19/04/2006	16/06/2006
Rostock	003	Mecklenburg-Vorpommern	19/04/2006	16/06/2006
Hannover	241	Niedersachsen	20/03/2006	16/06/2006
Göttingen	152	Niedersachsen	21/03/2006	16/06/2006
Essen	113	Nordrhein-Westfalen	16/03/2006	16/06/2006
Münster	515	Nordrhein-Westfalen	04/04/2006	16/06/2006
Aachen	313	Nordrhein-Westfalen	20/03/2006	16/06/2006
Köln	315	Nordrhein-Westfalen	20/03/2006	16/06/2006
Düsseldorf	111	Nordrhein-Westfalen	22/03/2006	16/06/2006
Bonn	314	Nordrhein-Westfalen	16/03/2006	16/06/2006
Mainz	315	Rheinland-Pfalz	16/03/2006	16/06/2006
Dresden	262	Sachsen	18/04/2006	16/06/2006
Leipzig	713	Sachsen	18/04/2006	16/06/2006
Magdeburg	003	Sachsen-Anhalt	21/03/2006	16/06/2006
Halle	202	Sachsen-Anhalt	16/03/2006	16/06/2006
Jena	053	Thüringen	24/03/2006	16/06/2006

NOTE.— This table shows the start and end date of the strike at each participating university hospital.

TABLE A.2. Physicians and health care workers in hospitals

Year	All Hospitals			University Hospitals		
	Total	Physicians	Other Medical	Total	Physicians	Other Medical
2000	1,100,471	122,062	978,409	172,867	24,398	148,469
2001	1,101,356	123,819	977,537	173,114	24,758	148,356
2002	1,112,421	126,047	986,374	174,850	25,084	149,766
2003	1,096,420	128,853	967,567	173,091	25,154	147,937
2004	1,071,846	129,817	942,029	168,980	25,171	143,809
2005	1,063,154	131,115	932,039	170,160	25,435	144,725
2006	1,064,377	133,649	930,728	171,895	25,781	146,114
2007	1,067,287	136,267	931,020	172,782	26,241	146,541
2008	1,078,212	139,294	938,918	173,182	26,488	146,694
2009	1,096,520	143,967	952,553	177,535	27,632	149,903
2010	1,112,959	148,696	964,263	181,954	28,443	153,511

NOTE.— Source: Information System of the Federal Health Monitoring.

TABLE A.3. Covariate balancing tests

	Observations	Mean	(1)	(2)	(3)
N Beds	395	1,130.48	432.597**	-22.986	-5.064
			(191.028)	(27.176)	(15.843)
S ICU Beds	395	0.08	0.020**	0.001	0.002
			(0.009)	(0.002)	(0.003)
N Physicians	395	573.50	337.988***	37.294	8.034
			(106.905)	(23.928)	(20.568)
S Female Physicians	395	0.33	-0.003	0.001	-0.004
			(0.017)	(0.008)	(0.011)
N Physicians in Charge	395	40.74	15.316	-2.302	-3.242
			(12.246)	(4.399)	(3.192)
N Assistant Physicians	395	433.12	227.567**	20.984	10.834
			(87.075)	(17.948)	(11.049)
N Non-Physicians	395	3,294.67	1972.984***	202.625*	44.069
			(560.346)	(108.468)	(104.998)
S Female Non-Physicians	395	0.77	-0.010	-0.008*	-0.002
			(0.013)	(0.004)	(0.003)
N Trainee Positions	395	352.22	134.706**	-20.573	-22.718
			(66.874)	(19.103)	(17.253)
N Childbirth	395	1,167.57	508.146	51.139	9.083
			(413.760)	(50.383)	(38.120)
S C-sections	395	0.28	0.119*	-0.005	-0.002
			(0.059)	(0.023)	(0.023)
Year FE			✓	✓	✓
Hospital FE				✓	✓
Hospital Specific Trends					✓

NOTE.— Robust standard errors clustered at the hospital level are reported in parenthesis. Significance levels: $*$ 0.10 $**$ 0.05 $***$ 0.01. Regressions successively include year fixed effects, hospital fixed effects and hospital specific time trends. In comparison to the main specification, the treatment period is defined as 2006, as data is only available on a yearly level. A zero share of C-sections is assumed for hospitals without maternity wards. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

TABLE A.4. Difference-in-differences estimates: Heterogeneity by Deferability

	QUARTILE 1			QUARTILE 2		
	Mean	(1)	(2)	Mean	(1)	(2)
Cases	167.82	-22.039***	-19.756***	166.81	-22.870***	-20.542***
		(4.660)	(2.735)		(4.848)	(2.758)
Log Cases	3.64	-0.160***	-0.154***	3.58	-0.173***	-0.167***
		(0.025)	(0.022)		(0.024)	(0.022)
Mortality Rate	1.53	0.186**	0.196**	1.52	0.313***	0.312***
		(0.091)	(0.092)		(0.089)	(0.090)
10 Day Mortality Rate	0.66	0.093	0.091	0.66	0.126**	0.121*
		(0.058)	(0.058)		(0.062)	(0.062)
Log Deaths	0.24	-0.000	-0.000	0.24	-0.034*	-0.028
		(0.018)	(0.017)		(0.020)	(0.017)
Length of Stay	7.48	0.279***	0.300***	8.79	0.477***	0.459***
		(0.098)	(0.087)		(0.084)	(0.081)
Surgery Rate	38.34	-0.398	-2.065*	42.73	0.184	-1.780
		(1.519)	(1.175)		(1.900)	(1.201)
	QUARTILE 3			QUARTILE 4		
	Mean	(1)	(2)	Mean	(1)	(2)
Cases	155.72	-20.668***	-18.715***	142.07	-18.800***	-16.517***
		(4.449)	(2.556)		(4.659)	(2.351)
Log Cases	3.49	-0.127***	-0.128***	3.41	-0.044***	-0.040***
		(0.024)	(0.018)		(0.015)	(0.010)
Mortality Rate	1.30	0.165***	0.168***	2.79	-0.007	-0.010
		(0.046)	(0.043)		(0.095)	(0.092)
10 Day Mortality Rate	0.69	0.041	0.040	1.91	0.001	-0.006
		(0.037)	(0.036)		(0.072)	(0.072)
Log Deaths	0.18	-0.004	-0.001	0.35	-0.021	-0.017
		(0.015)	(0.014)		(0.018)	(0.013)
Length of Stay	9.05	0.407***	0.383***	7.55	0.116	0.118*
		(0.104)	(0.091)		(0.075)	(0.063)
Surgery Rate	35.29	0.493	-1.161	27.83	1.030	-0.800
		(1.426)	(0.884)		(1.252)	(0.806)
Year FE		✓	✓		✓	✓
Hospital FE		✓	✓		✓	✓
Hospital Specific Trends			✓			✓

NOTE.— Robust standard errors clustered at the hospital level are reported in parentheses. Significance levels: $*$ 0.10, $**$ 0.05, $***$ 0.01. Regressions include day of week, week, year, and hospital fixed effects, and hospital-specific time trends where indicated. Regressions are weighted by the number of pre-strike admissions. Deferability is defined following Card, Dobkin and Maestas (2009) and Halla, Liu and Liu (2019). Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

TABLE A.5. Difference-in-differences estimates: Placebo regressions for 2004

	Observations	Mean	(1)	(2)
Cases	143,744	158.32	-1.032	-1.311
			(4.081)	(4.265)
Log Cases	143,744	4.90	0.009	0.007
			(0.014)	(0.015)
Mortality Rate	143,744	1.75	0.021	-0.006
			(0.058)	(0.038)
10-day Mortality Rate	143,744	0.96	0.002	-0.015
			(0.037)	(0.025)
Number of Deaths	143,744	2.46	0.002	-0.034
			(0.110)	(0.101)
Length of Stay	143,744	8.17	-0.021	-0.026
			(0.061)	(0.062)
Surgery Rate	143,744	35.95	2.610	2.395
			(1.853)	(1.865)
ACS Rate	143,744	20.30	0.056	0.077
			(0.178)	(0.180)
Emergency Rate	143,156	28.41	-0.121	-0.106
			(0.204)	(0.202)
Urgency Indicator	143,156	0.34	-0.002	-0.001
			(0.002)	(0.002)
Deferability Index	143,549	0.12	-0.000	-0.000
			(0.000)	(0.000)
Year FE			✓	✓
Hospital FE			✓	✓
Hospital Specific Trends				✓

NOTE.— Robust standard errors clustered at the hospital level are reported in parenthesis. Significance levels: $*$ 0.10 $**$ 0.05 $***$ 0.01. Regressions include day of week, week, year, and hospital fixed effects, and hospital specific time trends. Regressions are weighted by number of pre-strike admissions. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

TABLE A.6. Difference-in-differences estimates: Alternative Control Group

	Observations	Mean	(1)	(2)
Cases	267,382	161.83	-21.977***	-19.536***
			(3.718)	(2.354)
Log Cases	267,382	4.72	-0.138***	-0.125***
			(0.016)	(0.013)
Mortality Rate	267,382	2.07	0.270***	0.199***
			(0.045)	(0.041)
10-day Mortality Rate	267,382	1.19	0.135***	0.095***
			(0.036)	(0.034)
Number of Deaths	267,382	3.07	0.006	-0.035
			(0.090)	(0.055)
Length of Stay	267,382	8.28	0.392***	0.261***
			(0.064)	(0.053)
Surgery Rate	267,382	37.35	-1.765	-3.004**
			(1.607)	(1.170)
ACS Rate	267,382	21.93	-0.397*	-0.496***
			(0.216)	(0.175)
Emergency Rate	84,648	27.65	1.494***	1.484***
			(0.350)	(0.353)
Urgency Indicator	266,644	0.37	0.017***	0.014***
			(0.003)	(0.002)
Deferability Index	266,935	0.13	0.004***	0.004***
			(0.001)	(0.001)
Year FE			✓	✓
Hospital FE			✓	✓
Hospital Specific Trends				✓

NOTE.— Robust standard errors clustered at the hospital level are reported in parenthesis. Significance levels: $*$ 0.10 $**$ 0.05 $***$ 0.01. Regressions include day of week, week, year, and hospital fixed effects, and hospital specific time trends. Regressions are weighted by number of pre-strike admissions. The alternative control group consists of all hospitals in Berlin and Hesse. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

TABLE A.7. Difference-in-differences estimates: Alternative Treatment Indicator

	Observations	Mean	(1)	(2)
Cases	143,744	158.32	-20.065***	-17.453***
			4.984	2.645
Log Cases	143,744	4.90	-0.107***	-0.103***
			0.019	0.014
Mortality Rate	143,744	1.75	0.128***	0.134***
			0.044	0.042
10-day Mortality Rate	143,744	0.96	0.055	0.054
			0.036	0.035
Number of Death	143,744	2.46	-0.097	-0.064
			0.095	0.062
Length of Stay	143,744	8.17	0.226***	0.217***
			0.058	0.054
Surgery Rate	143,744	35.95	-0.473	-1.856*
			1.510	0.945
ACS Rate	143,744	20.30	0.009	-0.134
			0.233	0.193
Emergency Rate	143,156	28.41	1.364***	1.312***
			0.334	0.329
Urgency Indicator	143,156	0.34	0.010***	0.010***
			0.002	0.002
Deferability Index	143,549	0.12	0.003***	0.003***
			0.001	0.001
Year FE			✓	✓
Hospital FE			✓	✓
Hospital Specific Trends				✓

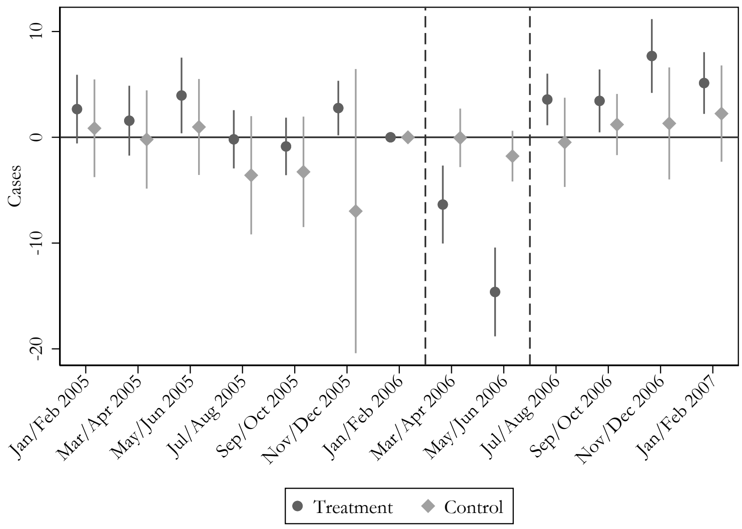
NOTE.— Robust standard errors clustered at the hospital level are reported in parenthesis. Significance levels: $*$ 0.10 $**$ 0.05 $***$ 0.01. Regressions include day of week, week, year, and hospital fixed effects, and hospital specific time trends. Regressions are weighted by number of pre-strike admissions. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

TABLE A.8. Pre-trend Test

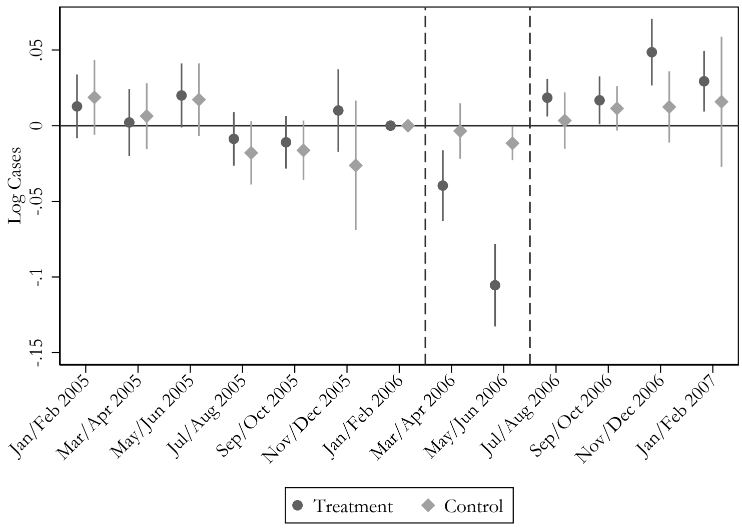
	(1) Log Cases	(2) Mortality Rate	(3) 10-Day Mortality Rate	(4) Cases
Trend × Treated	-0.0000262	-0.0000480	-0.0000434	-0.00972
	(0.0000325)	(0.0000448)	(0.0000264)	(0.0106)

NOTE.— Robust standard errors clustered at the hospital level are reported in parenthesis. Significance levels: $*$ 0.10 $**$ 0.05 $***$ 0.01. Regressions are weighted by number of pre-strike admissions. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

FIGURE A.1. Admissions - by Treatment Status



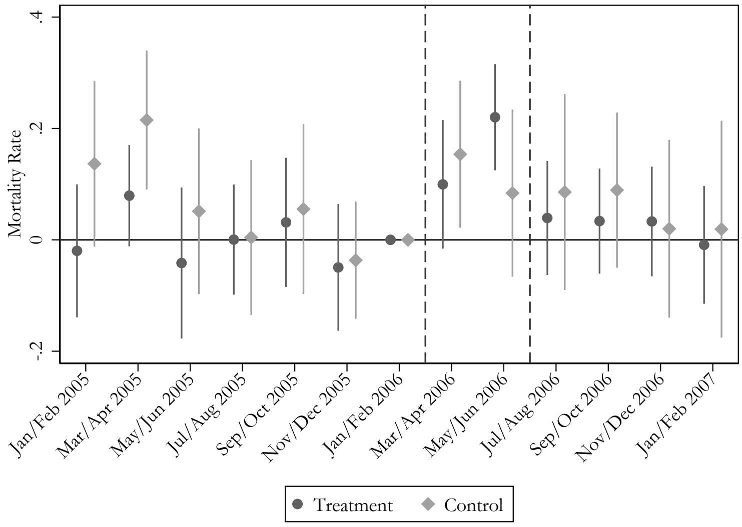
(a) Cases



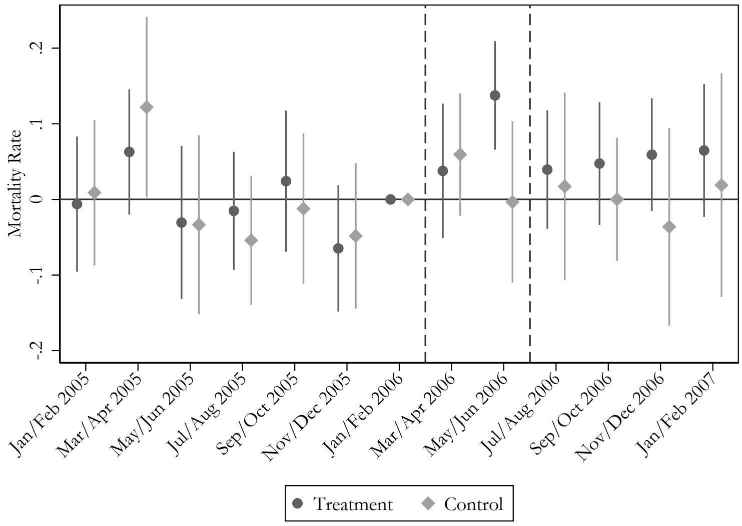
(b) Log Cases

NOTE.— The figures show event study estimates for cases and log cases by treatment status, controlling for day of week, week, and hospital fixed effects. The vertical lines represent the beginning and end of the strike. 95% confidence intervals included. Robust standard errors are clustered at the hospital level. Regressions are weighted by number of pre-strike admissions. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

FIGURE A.2. Mortality Rate - by Treatment Status



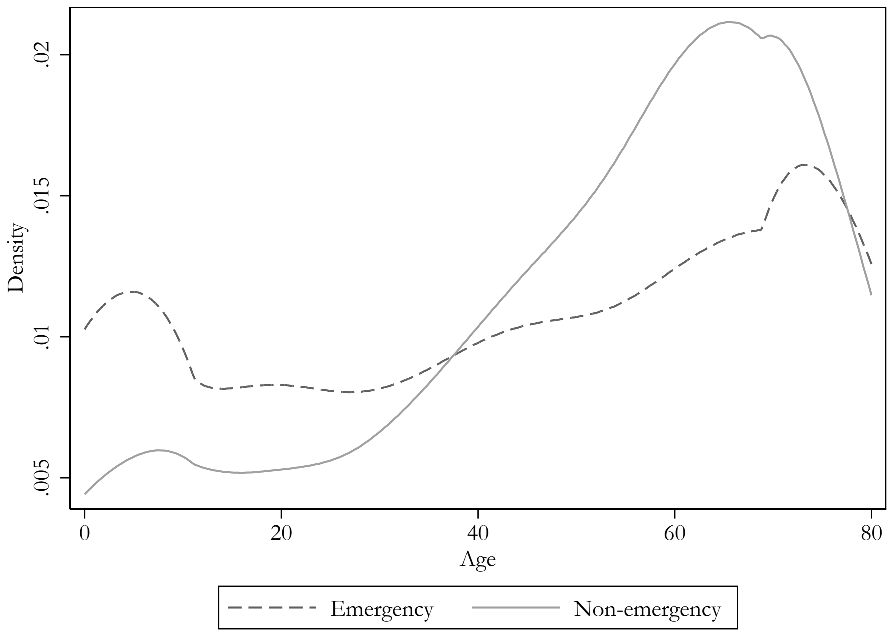
(a) In-hospital Mortality Rate



(b) 10-Day In-hospital Mortality Rate

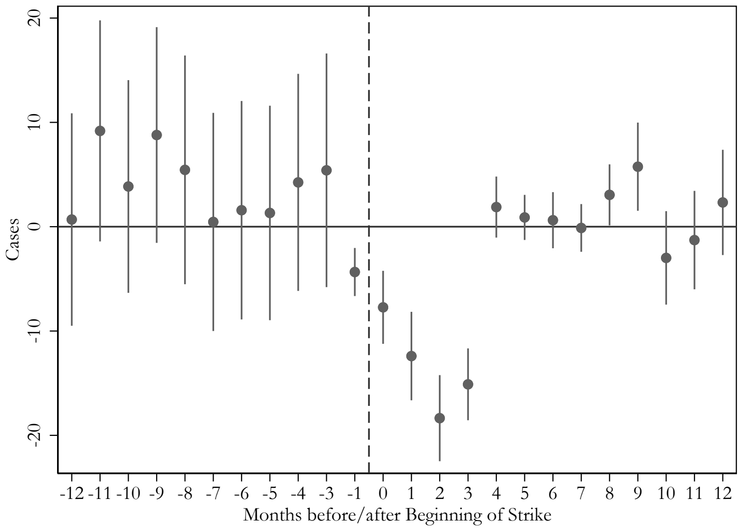
NOTE.— The figures show event study estimates for mortality rate and 10-day mortality rate by treatment status, controlling for day of week, week, and hospital fixed effects. The vertical lines represent the beginning and end of the strike. 95% confidence intervals included. Robust standard errors are clustered at the hospital level. Regressions are weighted by number of pre-strike admissions. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

FIGURE A.3. Age Distribution by Emergency Status

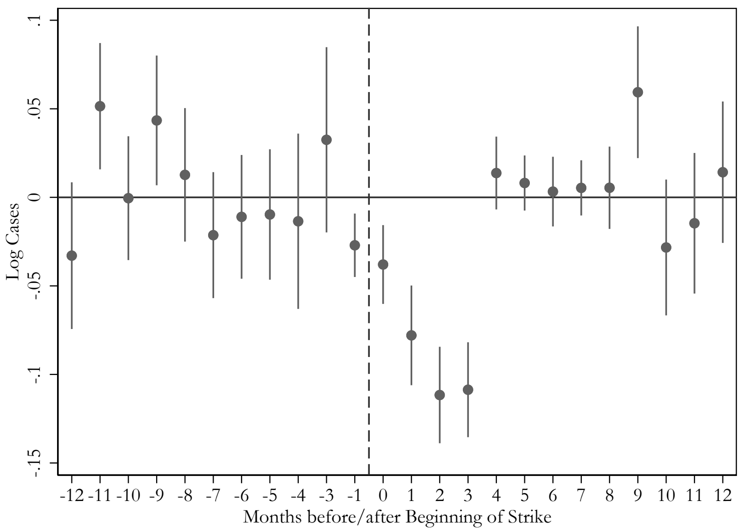


NOTE.— The figure shows the age distribution by emergency status. Based on individual level hospital data. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

FIGURE A.4. Alternative Estimator - Admissions



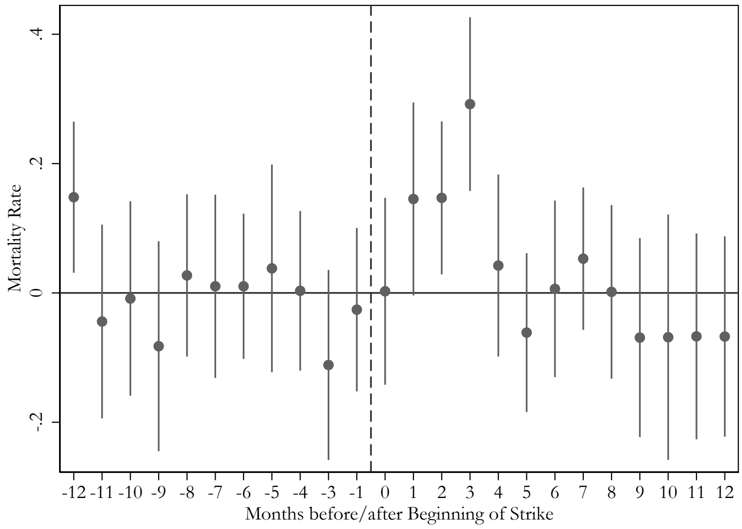
(a) Cases



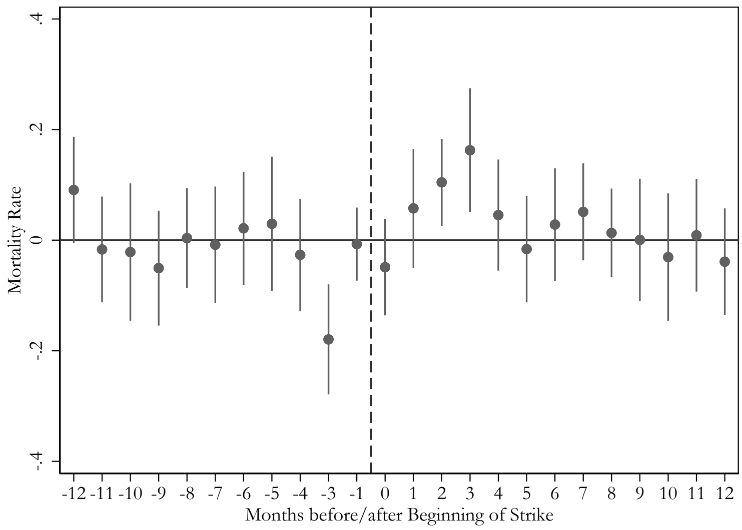
(b) Log Cases

NOTE.— The figures show event study estimates for cases and log cases, controlling for day of week, week, and hospital fixed effects using the estimator of Sun and Abraham (2021). Reference period $t-2$. The vertical lines represent the beginning and end of the strike. 95% confidence intervals included. Robust standard errors are clustered at the hospital level. Regressions are weighted by number of pre-strike admissions. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

FIGURE A.5. Alternative Estimator - Mortality Rate



(a) In-hospital Mortality Rate



(b) 10-Day Mortality Rate

NOTE.— The figures show event study estimates for mortality rate and 10-day mortality rate, controlling for day of week, week, and hospital fixed effects using the estimator of Sun and Abraham (2021). Reference period $t-2$. The vertical lines represent the beginning and end of the strike. 95% confidence intervals included. Robust standard errors are clustered at the hospital level. Regressions are weighted by number of pre-strike admissions. Source: FDZ der Statistischen Ämter des Bundes und der Länder (2000-2008*a*), own calculations.

Appendix B Variable Definitions Control and Outcome

Treatment Variable: Dummy variable taking on the value 1 for admissions to a striking hospital during the strike period.

Female: Dummy variable taking on the value 1 for females.

Age: Age in years, continuous variable.

Married: Dummy variable taking on the value 1 for married individuals.

Cases: Number of admissions per day per hospital.

Log Cases: Logarithmised number of admissions per day per hospital.

Length of Stay: Length of hospital stay in days.

Surgery Rate: Surgeries per 100 admissions.

Mortality Rate: Deaths per 100 admissions.

10 Day Mortality Rate: Deaths within 10 days after admission per 100 admissions.

District Mortality Rate: Deaths per 10,000 inhabitants.

ACS Rate: Ambulatory care sensitive admissions based on Sundmacher et al. (2015) per 100 admissions.

Urgency Indicator: Urgency indicator based on Krämer, Schreyögg and Busse (2019), continuous variable taking on values between 0 and 1.

Emergency Rate: Admissions indicated as emergency based on Krämer, Schreyögg and Busse (2019) per 100 admissions. Derived from urgency indicator. Admission is defined as an emergency admission if urgency indicator is above 0.5.

Deferability Index: Fraction of weekend admissions relative to weekday admissions per 3-digit ICD-diagnosis (see Card, Dobkin and Maestas (2009); Halla, Liu and Liu (2019)), continuous variable taking on values between 0 and 1.

Log Deaths: Logarithmised number of deaths per day per district.